Reference-Coupled Erasure: the Lookahead Discount

A problem statement (GO-15)

Geometric Observation program

Statement v0.2 — August 2026 (Theorem R$^+$ + DC form; convexity refuted; the $\mathcal{F}_{\Delta c}$ face certified; probe anchors corrected at matched floor)

Abstract

GO-14's verification pass discovered that its own U-independence hypothesis fences off a different object: records permitted to correlate with the noise of the eraser's aging reference erase at a fraction of the reference-independent cost (probe: 0.079 vs 0.567 bits/symbol at $\Delta = 0$, $n = 16$, $D = 0.3$ — an 86% collapse). This statement promotes that object to its own campaign. The probe's operational verdict frames it: the collapse is gated by *encoder lookahead past the eraser's access horizon*, and the realistic coordinate is the S-constructible family (the record writer reads the aging context), with general U-access a labeled fictitious relaxation. Nothing here is a claim; every probe number is a class-conditional upper bound; the $\mathcal{F}_{\Delta c}$ family now has a certified two-sided face, the general one does not.

1 Setting

As in GO-14: V AR(1), pole a; $Y = \rho V + N$; reference $S = V + U$, $\mathrm{Var}(U) = \tau^2$; per-cell erasure of $\hat{Y}^n$ against $(S^{t+\Delta}, \hat{Y}^{t-1})$; coordinate $L_a(\Delta)$. Families: $\mathcal{F}_0$ (U-independent, GO-14's), $\mathcal{F}_{\Delta c}$ (Δ-lag-causal U-coupling, $A_u[t,s] = 0$ for $s > t + \Delta$ — the chain-rule-preserving class), $\mathcal{F}_S$ ($\hat{Y} = A_y Y + A_s S + Z$: the encoder reads the reference, with an explicit lookahead-window parameter), and $\mathcal{F}_U$ (general U-access — fictitious: it resolves S into $V + U$).

Proposition 1 (Corrected identity — from the GO-14 R-IND-5 record, re-verified at 24 probe optimizers)**.** *$nL_a = \text{block} + C_\Delta - \sum_t I(S_{>t+\Delta}; \hat{Y}_t \mid T, S^{t+\Delta}, \hat{Y}^{t-1})$; the correction vanishes iff the record is Δ-lag causally U-coupled. At probe optimizers of $\mathcal{F}_{\Delta c}$ the correction is 0 to 10^{-16}; at the $\mathcal{F}_U$ winner it carries $74\times$ the final value (only the difference of the diverging terms is physical near the deterministic-record boundary).*

Conjecture 1 (Strict lookahead hierarchy)**.** *For all $n \geq 8$, $\Delta \leq n - 2$, and any noise floor $\sigma_0^2 \leq 10^{-2}$: $\min_{\mathcal{F}_U} < \min_{\mathcal{F}_S} < \min_{\mathcal{F}_{\Delta c}} < \min_{\mathcal{F}_0}$, with the Δ-causal optimum essentially causally-S-constructible ($A_u \approx A_v$ measured) — within-horizon U-coupling adds nothing beyond accessible S. Probe anchors at $(16, 0)$, floorless: $0.079 < 0.290 < 0.522 < 0.567$.* ***Corrected at matched floor $\sigma_0^2 = 10^{-6}$ (LB-face prover, 2026-08-07):*** *$0.104 < 0.170 < 0.522 < 0.567$ — the probe's $\mathcal{F}_S$ search was under-converged, so lookahead captures 85.8% of the collapse,* not 57%, *and direct U-resolution is essential for only ≈0.066 bits. The qualitative verdict (collapse iff the encoder's reference window exceeds the eraser's horizon) is unchanged.* ***Two levels are now theorems:*** *$\min \mathcal{F}_S \leq 0.1698616 < 0.522354033 = \mathrm{LB}(\mathcal{F}_{\Delta c})$, and $\min \mathcal{F}_{\Delta c} < \min \mathcal{F}_0$ at four cells; $\min \mathcal{F}_U < \min \mathcal{F}_S$ stays conjectural.*

Conjecture 2 (Horizon-matched anti-coupling)**.** *Optimizers concentrate* $\geq 60\%$ *of their* U*-coupling mass, negatively, on the single first-beyond-horizon band* $s = t + \Delta + 1$ *(measured* 63–93% *across all probe winners;* $\operatorname{corr}(\hat{Y}_t, U_{t+1}) \approx -0.245$*): the record* anticipates the first reference sample the eraser cannot see — *GO-14's horizon-matched mechanism transplanted from* V*-cancellation to* U*-anticipation, and a cross-cell code effect (a lone band on a scalar record recovers almost none of it).*

Conjecture 3 (Δ-reversal)**.** $\min_{\mathcal{F}_U} L_a(\Delta)$ *is strictly increasing in* Δ*, rejoining the block coordinate at* $\Delta = n$ *— the lag knob reverses sign across the* U*-coupling boundary (the* $\mathcal{F}_0$ *coordinate strictly decreases; probe:* $0.079 \to 0.201 \to 0.219$ *vs* $0.567 \to 0.536 \to 0.535$ *at* $n = 16$*). The* $\mathcal{F}_S$ *shape is open.*

Conjecture 4 (Boundary attainment and the floor law)**.** *At* $\sigma_0 = 0$ *the infimum over* $\mathcal{F}_U$ *is attained only on the singular (deterministic-record) closure;* $\sigma_0 \mapsto \min L_a$ *is continuous and increasing with the floor constraint active (probe:* $0.079/0.105/0.149/0.366$ *at* $\sigma_0^2 = 10^{-8}/10^{-6}/10^{-4}/10^{-2}$*, all below* $\mathrm{block}_{16} = 0.535$ *— the phenomenon survives realistic noise; the* value *is floor-relative and every claim must carry its floor). The* $\sigma_0 \to 0$ *limit at fixed* n *is open.*

Proposition 2 (Theorem R$^+$ and the DC form — LB-face prover 2026-08-07, R-IND-5 pending)**.** *In extended coordinates* (H_w, H_u, Γ) *on the LMI cone* $N := \Gamma - G P_e G' \succeq 0$ *(with* $N = \operatorname{Cov}(Z)$*, and the distortion affine in* (H_w, Γ) *and* independent of H_u*), the representation holds for every record and every nondecreasing schedule, and splits as* $nL_a = L_a^+ - \mathrm{Dterm}$ *where* L_a^+ *is GO-14's coordinate for the* augmented *source* $W^+ = (W, U)$ *— an exact DC decomposition with* both *pieces convex in closed form.* Dterm $\equiv 0$ *iff the* U*-coupling is* Δ*-lag causal. This also repairs GO-14's moment-form remark: the* extended *form is exact at every* A_u*, causal or not.*

Proposition 3 (Non-convexity — REFUTED with a canonical counterexample)**.** nL_a *is* not *convex on the extended cone. Base point: the GO-14 certified* $\mathcal{F}_0$ *optimizer at* $(16, 0)$*; since the distortion is* H_u*-free,* $\pm\varepsilon$ *on the* U*-block gives two feasible records at identical distortion whose exact midpoint is that optimizer, with Jensen violation* $+1.178143 \times 10^{-1}$ *bits (reproduced independently). Restricted Hessians at that point: all directions* -3.85*;* $\mathcal{F}_0$ *only* $+0.261$*;* $\mathcal{F}_{\Delta c}$ *only* $+0.260$*; pure-*H_u -2.358*. The defect is confined to lookahead-*U *directions and is carried entirely by the concave* Dterm *— so the obstacle is the* denominator's S*-side pivot sum, not the numerator regrouping, and no regrouping removes it.*

Question 1 (A lower bound for the headline cell)**.** *For* $\mathcal{F}_U$ *and* $\mathcal{F}_S$ *the best certified lower bound remains* 0*: duality needs convexity (refuted), and DC decoupling, a numerator-floor minorant, and an information-inequality relaxation are all vacuous — at the optimum the value sits inside a* $62\times$ *cancellation, and every decoupling loses* $\approx \frac{1}{2}\log_2(1/\sigma_0^2)$ *bits. The* $\mathcal{F}_{\Delta c}$ *family, by contrast, has a certified two-sided face (its section is linear and* Dterm *vanishes there):* $[0.522354033, 0.522356390]$ *at* $(16, 0)$*, the first lower bound for any* U*-coupled family. Remaining route for the headline cell: spatial branch-and-bound on the* H_u *block over the DC form — after equal-depth matched-floor reruns, since the present* $\mathcal{F}_S/\mathcal{F}_U$ *anchors are optimization-depth artefacts at the second decimal. No* n*-trend may be claimed meanwhile.*

Question 2 (The Landauer face)**.** *The erasure cost of a record against a causally-growing reference is* not encoder-independent*: it is gated by how far past the eraser's horizon the record-writer could read. What reset protocol realizes the lookahead discount, and what does the discount cost at write time? (The GO-9 coordinated-reset lane is the in-house exterior.)*

House rules: novelty flank swept 2026-08-06 (claimable; citation spine in the probe record — del Rio, Sagawa–Ueda, Berta et al., Cover–Chiang, Gelfand–Pinsker, arXiv:1507.04924, Ito–Sagawa, Cuff et al., Perarnau-Llobet et al.; two targeted re-sweeps owed pre-seal). R-IND-5 before assertion; §5.1 at every seal; the $\mathcal{F}_U$ family is always labeled fictitious next to any headline number.